

Special Triangle Congruence Theorems and Related Theorems

Jonathan R. Bacolod

Right Triangle Congruence Theorems

provide shortcuts for proving congruence of right triangles.

These include

HL, LL, HA, LA

HL and LL Congruence

HL (Hypotenuse-Leg) states that two right triangles are congruent if their hypotenuses and one pair of corresponding legs are congruent.

HL

LL (Leg-Leg) states that two right triangles are congruent if their two pairs of corresponding legs are congruent.

LL

HA and LA Congruence

HA (Hypotenuse-Acute Angle) states that two right triangles are congruent if their hypotenuses and one pair of corresponding acute angles are congruent.

HA

LA (Leg-Acute Angle) states that two right triangles are congruent if one pair of corresponding legs and one pair of corresponding acute angles are congruent.

LA

Isosceles Triangle Theorem

The **Isosceles Triangle Theorem** states that if two sides of a triangle are congruent, then the angles opposite those sides are congruent.

$$PQ = PR$$

Therefore,

$$\angle Q = \angle R$$

Converse of Isosceles Triangle Theorem

The **Converse of the Isosceles Triangle Theorem** states that if two angles of a triangle are congruent, then the sides opposite those angles are congruent.

$$\angle X = \angle Z$$

Therefore,

$$YZ = XY$$

Perpendicular Bisector Theorem

The **Perpendicular Bisector Theorem** states that any point on the perpendicular bisector of a segment is equidistant from the endpoints of the segment.

P lies on the perpendicular bisector of AB

Therefore,

$$PA = PB$$

Converse of Perpendicular Bisector

The **Converse of the Perpendicular Bisector Theorem** states that if a point is equidistant from the endpoints of a segment, then it lies on the perpendicular bisector of the segment.

$$QA = QB$$

Therefore, Q lies on the perpendicular bisector of AB .

Midline Theorem

The **Midline (Midsegment) Theorem** states that the segment joining the midpoints of two sides of a triangle is parallel to the third side and has one-half its length.

$$DE \parallel BC$$

and

$$DE = \frac{1}{2}BC$$

Examples

Example 1

Two right triangles have hypotenuses measuring 15 cm and corresponding legs measuring 9 cm.

Which right triangle congruence theorem proves the triangles congruent?

HL

Example 2

Right triangles $\triangle ABC$ and $\triangle DEF$ have congruent hypotenuses and one pair of corresponding acute angles congruent.

Identify the right triangle congruence theorem.

HA

Example 3

In isosceles $\triangle PQR$,

$$PQ = PR$$

If $\angle Q = 52^\circ$, find $\angle R$ and $\angle P$.

$$\angle R = 52^\circ$$

$$\angle P = 180^\circ - 52^\circ - 52^\circ$$

$$\boxed{\angle P = 76^\circ}$$

Example 4

In $\triangle XYZ$,

$$\angle X = \angle Z$$

If $XY = 13$ cm, determine YZ .

Since equal angles have congruent opposite sides,

$$YZ = XY$$

$$\boxed{YZ = 13 \text{ cm}}$$

Example 5

Segment AB has midpoint M .

Point P lies on the perpendicular bisector of AB .

If $PA = 18$ cm, find PB .

$$PA = PB$$

Therefore,

$$\boxed{PB = 18 \text{ cm}}$$

Example 6

Point Q is equidistant from the endpoints of segment CD .

What can you conclude about the location of Q ?

$$QC = QD$$

Therefore, Q lies on the perpendicular bisector of CD .

Example 7

In $\triangle ABC$, D and E are the midpoints of AB and AC .

If $BC = 24$ cm, find DE .

$$DE = \frac{1}{2}BC$$

$$DE = \frac{1}{2}(24)$$

$$\boxed{DE = 12 \text{ cm}}$$

Example 8

In $\triangle LMN$, XY joins the midpoints of LM and LN .

If $XY = 11$ cm, determine MN .

$$XY = \frac{1}{2}MN$$

$$11 = \frac{1}{2}MN$$

$$\boxed{MN = 22 \text{ cm}}$$

Example 9

A triangle has sides 14, 14, and 20.

Determine the measures of the two base angles if the vertex angle measures 68° .

Since the two sides are congruent, the base angles are congruent.

$$\angle B = \angle C$$

$$\angle B = \angle C = \frac{180^\circ - 68^\circ}{2}$$

$$\boxed{\angle B = \angle C = 56^\circ}$$

Practice Exercises

Practice 1

Two right triangles have hypotenuses measuring 15 cm and corresponding legs measuring 9 cm.

Which right triangle congruence theorem proves the triangles congruent?

Practice 2

Right triangles $\triangle ABC$ and $\triangle DEF$ have congruent hypotenuses and one pair of corresponding acute angles congruent.

Identify the right triangle congruence theorem that applies.

Practice 3

In isosceles $\triangle PQR$,

$$PQ = PR$$

If $\angle Q = 52^\circ$, find $\angle R$ and $\angle P$.

Practice 4

In $\triangle XYZ$,

$$\angle X = \angle Z$$

If $XY = 13$ cm, determine the length of YZ .

Practice 5

Segment AB has midpoint M .

Point P lies on the perpendicular bisector of AB .

If $PA = 18$ cm, find PB .

Practice 6

Point Q is equidistant from the endpoints of segment CD .

What can you conclude about the location of Q ?

Practice 7

In $\triangle ABC$, D and E are the midpoints of AB and AC , respectively.

If $BC = 24$ cm, find DE .

Practice 8

In $\triangle LMN$, XY joins the midpoints of LM and LN .

If $XY = 11$ cm, determine MN .

Practice 9

A triangle has sides 14, 14, and 20.

Determine the measures of the two base angles if the vertex angle measures 68° .

Activities

Activity 1

Two right triangles have hypotenuses measuring 17 cm and corresponding legs measuring 8 cm.

Which right triangle congruence theorem proves the triangles congruent?

Activity 2

Right triangles $\triangle GHI$ and $\triangle JKL$ have congruent hypotenuses and one pair of corresponding acute angles congruent.

Identify the right triangle congruence theorem that applies.

Activity 3

In isosceles $\triangle ABC$,

$$AB = AC$$

If $\angle B = 61^\circ$, find $\angle C$ and $\angle A$.

Activity 4

In $\triangle DEF$,

$$\angle D = \angle F$$

If $DE = 18$ cm, determine the length of EF .

Activity 5

Segment XY has midpoint N .

Point R lies on the perpendicular bisector of XY .

If $RX = 21$ cm, find RY .

Activity 6

Point T is equidistant from the endpoints of segment MN .

What can you conclude about the location of T ?

Activity 7

In $\triangle PQR$, S and T are the midpoints of PQ and PR , respectively.

If $QR = 30$ cm, find ST .

Activity 8

In $\triangle ABC$, DE joins the midpoints of AB and AC .

If $DE = 16$ cm, determine BC .

Activity 9

A triangle has sides 16, 16, and 24.

Determine the measures of the two base angles if the vertex angle measures 74° .

Thank you!